

AI Study Buddy – Personalized Learning Assistant

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Abstract

AI Study Buddy is an AI-powered educational assistant designed to provide personalized study plans, AI tutoring, quiz generation, and learning recommendations. The project aligns with UN Sustainable Development Goal 4 (Quality Education) by improving access to personalized learning through modern AI technologies.

Introduction

Students often face challenges in organizing studies, identifying relevant learning resources, and receiving personalized academic guidance. AI Study Buddy addresses these issues using Artificial Intelligence.

Problem Statement

Traditional learning platforms provide the same content to every learner regardless of individual needs. Students require personalized study plans, instant doubt resolution, and adaptive learning support.

Objectives

- Create personalized study plans.
- Provide AI-powered academic assistance.
- Generate quizzes automatically.
- Recommend learning resources.
- Improve learning outcomes.

Technologies Used

- Retrieval-Augmented Generation (RAG)
- IBM Granite Models
- Agentic AI Concepts
- Natural Language Processing (NLP)
- IBM BOB
- Python
- Streamlit

System Architecture

User → Streamlit Interface → Agentic AI Layer → IBM Granite Models + RAG Pipeline → Personalized Responses (Study Plan, AI Tutor, Quiz Generator, Learning Resources).

Key Features

- AI Tutor
- Personalized Study Planner
- Quiz Generator
- Learning Resource Recommendations
- Progress Tracking Dashboard

Expected Impact

The solution supports students with personalized learning, better time management, improved academic performance, and contributes to SDG 4 by making quality education more accessible.

Future Scope

Voice-based tutoring, mobile application, LMS integration, multilingual support, analytics dashboard, and real-time collaboration.

Conclusion

AI Study Buddy demonstrates how AI technologies such as RAG, IBM Granite, and Agentic AI can be applied to improve educational accessibility and personalization. This concept showcases the positive role of AI in supporting Sustainable Development Goal 4.

References

1. IBM SkillsBuild
2. IBM Granite Documentation
3. United Nations Sustainable Development Goals
4. 1M1B AI + Sustainability Virtual Internship